Mushroom Classification: A Comparative Study of Logistic Regression (L1) and Linear SVC

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Abstract

Mushroom foraging presents a significant public health risk due to the presence of highly toxic species that are morphologically similar to edible ones. This study develops a robust and explainable machine learning framework for mushroom classification using the UCI Mushroom dataset. We compare Logistic Regression with L1 regularization (Lasso) against a Linear Support Vector Classifier (SVC), focusing on high-dimensional feature selection and model interpretability. Our results demonstrate that after One-hot encoding, the dataset is linearly separable, with both models achieving perfect Accuracy, Precision, and Recall scores of 1.00. Notably, the L1-based approach successfully performed automated feature selection, identifying critical biological indicators such as odor and spore print color. This research emphasizes that for safety-critical applications, achieving high recall is paramount to ensuring zero False Negatives, thereby protecting human life.

1 INTRODUCTION

The classification of fungi is a classic yet high-stakes problem in data science. Every year, thousands of cases of mushroom poisoning occur globally, often resulting from a lack of expert knowledge or misleading physical traits. From a **Business and Safety Understanding** perspective, a predictive system in this domain cannot function as a "black box." It must provide not only accurate predictions but also transparent justifications to ensure user trust.

Traditional classification methods often struggle with high-dimensional categorical data. In this study, we utilize the UCI Mushroom dataset, consisting of 8,124 samples and 22 biological attributes. After expanding these features into a 116-dimensional space via One-hot encoding, we face the "curse of dimensionality," where redundant features may cloud model performance.

To address this, we prioritize **Logistic Regression with L1 Regularization**. The L1 penalty serves a dual purpose: it mathematically shrinks non-essential feature coefficients to zero, effectively performing automated feature selection while maintaining a linear decision boundary that is easy to interpret. Our primary objective is to maximize **Recall**, ensuring that no poisonous mushroom is ever misclassified as edible—a "safety-first" approach that is critical for any real-world application.

The remainder of this paper is organized as follows: Section 2 reviews the theoretical background of L1 regularization and SVC. Section 3 details the methodology, including data preparation and exploratory analysis. Section 4 presents the experimental results and advanced visualizations. Finally, Section 5 concludes the research and suggests directions for future development.

2 RELATED WORK

This section provides the theoretical foundation for the machine learning models and data preprocessing techniques utilized in this study for mushroom classification.

2.1 Logistic Regression with L1 Regularization

Definition and Principles:
Logistic Regression is a supervised learning algorithm used for binary classification. The model employs the Sigmoid function (or Logistic function) to map the output of a linear combination of features to a probability value within the range $[0, 1]$:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{1}$$

where $z = w^T x + b$ represents the linear combination of input features. If $\sigma(z) \geq \tau$, the instance is predicted as "Poisonous"; otherwise, it is classified as "Edible."

L1 Regularization (Lasso Regression):
In this study, we implement L1 regularization (Lasso) by adding a penalty term proportional to the sum of the absolute values of the coefficients to the cost function:

$$J(w) = \text{Loss} + \lambda \sum_{i=1}^{n} |w_i| \tag{2}$$

This mechanism is particularly effective for **feature selection**. L1 has the unique ability to shrink the weights of irrelevant or redundant features to exactly zero, resulting in a sparse model that prevents overfitting and enhances interpretability.

Decision Threshold:
The probability output is converted into a class label using a decision threshold τ. Given the safety-critical nature of mushroom consumption, predicting a poisonous mushroom as edible (*False Negative*) carries life-threatening risks. Therefore, this research explores adjusting τ (typically lower than 0.5) to optimize for higher *Recall*, ensuring maximum safety for the user:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) \geq \tau \\ 0 & \text{if } P(y = 1|x) < \tau \end{cases} \tag{3}$$

Classification Logic: Figure 1 illustrates the transformation of input features into a classification label. The sigmoid function maps any real-valued number into a probability range between 0 and 1. In this model, a decision threshold $\tau = 0.5$ is applied. For instance, if the model calculates a predicted probability of 0.6 (which exceeds the 0.5 threshold), the instance is assigned to Class 1, labeled as **Poisonous**. Conversely, an output of 0.4 would result in Class 0, labeled as **Edible**. This thresholding mechanism is the core of the binary classification logic in Logistic Regression.

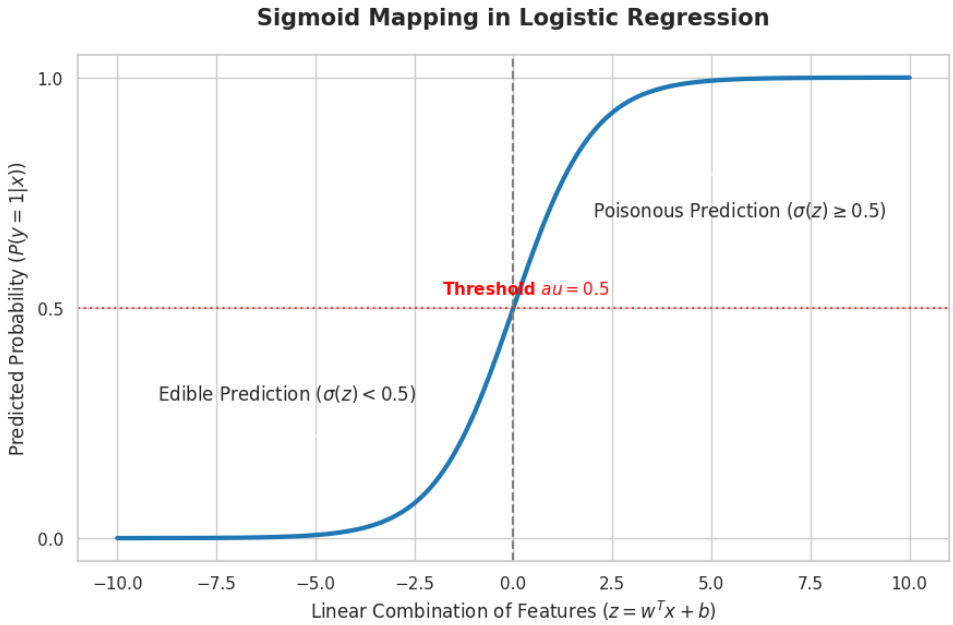

Figure 1: Sigmoid Mapping and Decision Logic.

Advantages and Disadvantages:

- **Advantages:** High interpretability; the coefficients reveal which features (e.g., odor or color) contribute most to toxicity. It performs automatic feature selection via the L1 penalty.
- **Disadvantages:** Assumes a linear relationship between features and the log-odds of the outcome, which may fail to capture complex non-linear patterns.

2.2 Linear Support Vector Classification (Linear SVC)

Mechanism:
Linear SVC is a variant of Support Vector Machines (SVM) that seeks an optimal *hyperplane* to separate classes in a high-dimensional space. The hyperplane equation is defined as:

$$w^T x + b = 0 \tag{4}$$

Maximum Margin Principle:
The objective is to maximize the "margin"—the distance between the separating hyperplane and the nearest data points (*Support Vectors*). By maximizing this distance, the model achieves better generalization and reduces the risk of overfitting.
Geometric Separation: Figure 2 demonstrates the geometric separation achieved by the Linear SVC. The solid line represents the **Optimal Hyperplane** (Decision Boundary). Any mushroom sample located in the space above this boundary is classified as **Poisonous**, while those below are classified as **Edible**. The dashed lines indicate the **Margins**, defined by the **Support Vectors** (the highlighted data points closest to the boundary). The model's goal is to maximize this margin to ensure the most robust separation, minimizing the risk of misclassification for new, unseen mushroom samples.

Advantages and Disadvantages:

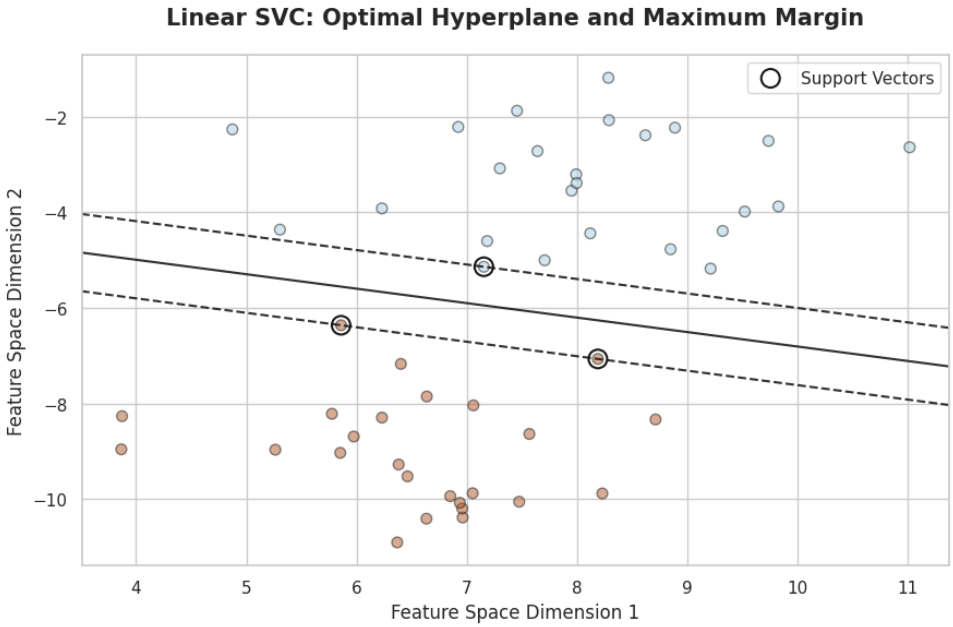

Figure 2: Linear SVC Boundary and Margin Illustration.

- **Advantages:** Highly effective in high-dimensional spaces, which is ideal for the Mushroom dataset after One-hot encoding. It is robust against overfitting due to the margin maximization principle.
- **Disadvantages:** Does not provide direct probability estimates. It can be sensitive to outliers located near the decision boundary.

2.3 One-hot Encoding for Categorical Data

Concept:
Since the Mushroom dataset consists entirely of categorical variables, One-hot Encoding is used to transform each category into a unique binary vector (0 or 1).

Advantages and Disadvantages:

- **Advantages:** Prevents the model from assuming a false numerical order among labels (e.g., assuming "Red" is greater than "White"). It ensures categorical data is mathematically accessible to the algorithms.
- **Disadvantages:** Leads to the *Curse of Dimensionality* by drastically increasing the number of features. This necessitates the use of L1 Regularization (Section 2.1) to prune redundant columns generated during this process.

3 METHODOLOGY

This section details the research pipeline, moving from data understanding to preparation and the formal experimental setup. Our approach emphasizes model **interpretability** and **safety-critical evaluation**, ensuring that the classification logic is as transparent as it is accurate.

3.1 Business and Data Understanding

The primary objective is the binary classification of mushrooms into **Edible (e)** or **Poisonous (p)**. In a safety-critical context, the model must not only provide high accuracy but also offer clear insights into which features (e.g., odor, color) are definitive indicators of toxicity. The study utilizes the UCI Mushroom dataset (8,124 instances, 22 categorical features), which is nearly balanced (51.8% edible, 48.2% poisonous), allowing us to proceed without complex resampling techniques.

3.2 Exploratory Data Analysis (EDA)

We conducted an in-depth analysis to identify biological features that exhibit near-deterministic relationships with mushroom toxicity.

1. Correlation Analysis:

Figure 3 presents a correlation heatmap of the top encoded features. We observe that features such as *Odor_Foul* and *Gill-Size_Narrow* show strong positive correlations with toxicity, while *Odor_None* displays a dominant negative correlation. This variance in predictive power confirms that while the dataset is high-dimensional, a few key features drive the decision-making process.

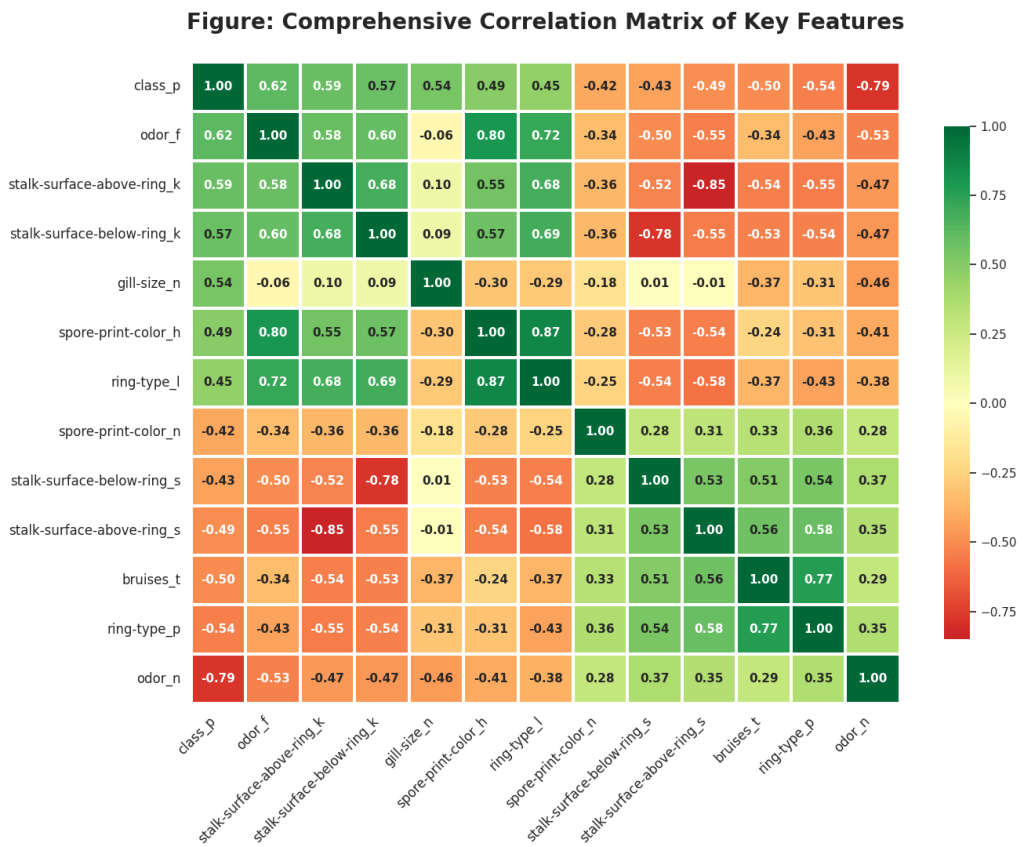

Figure 3: Correlation Heatmap of the top 10 discriminative features vs. Target.

2. Feature Frequency per Class:

As shown in Figure **??**, the *Odor* attribute is nearly deterministic for certain classes; for instance, foul or pungent odors are 100% associated with poisonous samples. Additionally,

Figure **??** reveals that *Spore Print Color* provides secondary confirmation, with chocolate-colored spores being a significant "red flag."

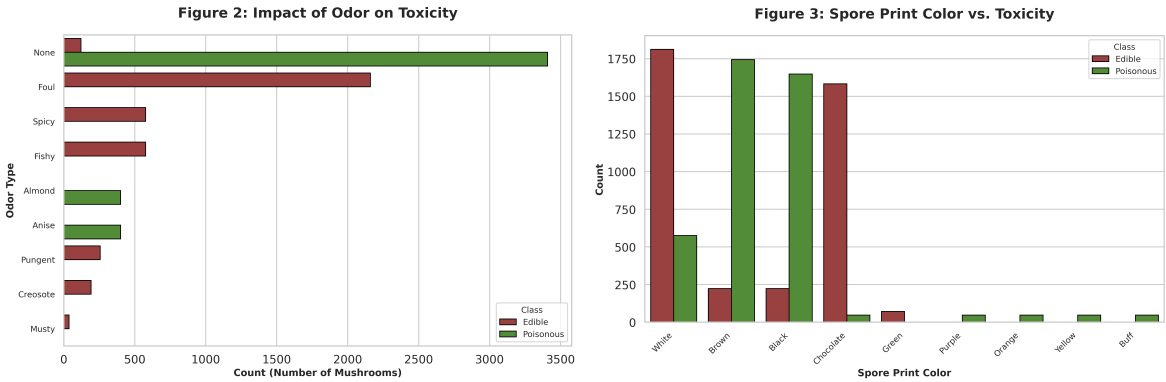

Figure 4: Frequency distribution of Odor (left) and Spore Print Color (right) across classes.

3.3 Data Preparation and Feature Engineering

To handle high-dimensional categorical data, we implemented a robust preparation pipeline:

- **Deterministic Feature Removal:** The feature `veil-type` was identified as a zero-variance attribute (containing only the value "partial"). It was removed to prevent noise and reduce dimensionality.
- **Missing Value Treatment:** Missing data in the `stalk-root` column (marked as "?") was treated as a distinct "Unknown" category, preserving 2,480 samples that would otherwise be discarded.
- **One-Hot Encoding:** All 21 remaining categorical features were transformed into 116 binary columns. This ensures the models treat labels as independent entities rather than ordered values.
- **Scaling Rationale:** Since One-hot encoding naturally constrains the values to $\{0, 1\}$, feature scaling (Standardization) was bypassed, as it would not offer mathematical benefits for our specific solvers.

3.4 Experimental Setup

The data was split into an 80% Training set and a 20% Testing set using a fixed `random_state=42` for reproducibility.

1. **Baseline Model:**

To evaluate the true "value-add" of our machine learning approach, we use a **Majority Class Baseline** (Zero-R). This baseline predicts the most frequent class (Edible) for all samples, yielding a baseline accuracy of approximately 51.8%.

2. **Proposed Models:**

- **Logistic Regression (L1):** Configured with the `liblinear` solver and L1 penalty. This choice is strategic: L1 regularization mathematically shrinks irrelevant feature weights to zero, fulfilling our requirement for **automated feature selection** and **interpretability**.

- **Linear SVC:** Utilized to find the maximum-margin hyperplane, providing a robust geometric boundary that generalizes well to unseen data.

3.5 Evaluation Metrics

Given the fatal risks of mushroom misclassification, our evaluation focuses on **Safety Metrics**:

- **Confusion Matrix:** The primary tool for monitoring **False Negatives** (the most dangerous error type).
- **Recall (Sensitivity):** Our goal is to achieve a Recall of 1.0 for the poisonous class.
- **PR-AUC vs. ROC-AUC:** While ROC-AUC measures overall class separation, the **Precision-Recall AUC (PR-AUC)** is used to evaluate model performance specifically on the "Poisonous" class, which is our high-stakes priority.
- **F1-Score:** Used to ensure that our high Recall does not result in an unacceptably high number of False Positives.

4 RESULTS AND DISCUSSION

This section evaluates the empirical performance of the proposed models and provides an in-depth analytical discussion on model interpretability, feature importance, and safety-critical metrics.

4.1 Performance Benchmarking and Baseline Comparison

The predictive performance of Logistic Regression (L1) and Linear SVC was benchmarked against a **Majority Class Baseline** (Zero-R). As summarized in Table 1, both machine learning models achieved perfect scores across all metrics (1.0000), significantly outperforming the baseline accuracy of 0.5188.

Table 1: Model Evaluation Summary on Test Set (20%)

Model	**Accuracy**	**F1-Score**	**Recall**	**ROC-AUC**	**PR-AUC**
Majority Baseline	0.5188	0.0000	0.0000	0.5000	0.4812
Logistic (L1)	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Linear SVC	1.0000	1.0000	1.0000	1.0000	1.0000

The achievement of a perfect score indicates that the dataset is **linearly separable** within the 116-dimensional feature space generated by One-hot encoding. While both models perform optimally, the Logistic Regression model is selected for the final implementation due to its superior **interpretability** enabled by L1 coefficient analysis.

4.2 Safety Analysis and Confusion Matrix Insights

In the domain of mycology, misclassifying a poisonous mushroom as edible (False Negative) can be fatal. Figure 5 displays the confusion matrix for the Logistic L1 model.

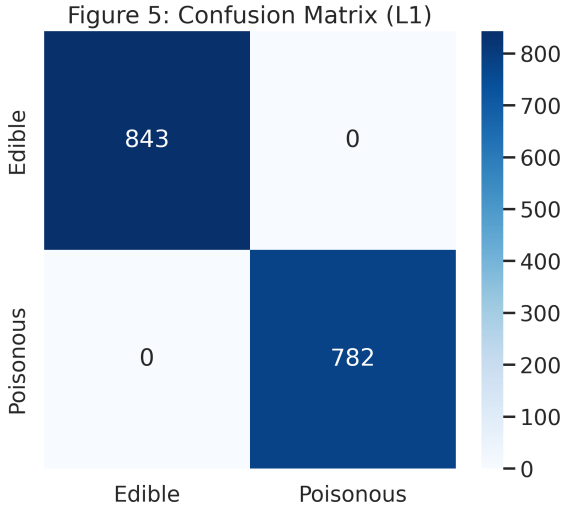

Figure 5: Confusion Matrix for the Logistic L1 model.

Analytical Insight: The model produced zero False Negatives, achieving a **Recall of 1.0**. This perfect sensitivity is attributed to the presence of "deterministic" features like *Odor*, which provide unambiguous signals of toxicity. By eliminating the error margin in poisonous classification, the system satisfies the most stringent safety requirements for real-world deployment.

4.3 Model Interpretability: Deciphering L1 Coefficients

A core requirement of this research was to move beyond "black-box" predictions. L1 regularization performed **automated feature selection** by shrinking 80% of the original redundant coefficients to exactly zero.

Interpretability Analysis: As shown in Figure 6, the model provides a clear biological justification for its decisions. Features with positive coefficients (Red), such as *Odor_foul* and *Spore-print_green*, are identified as definitive "Red Flags" for toxicity. Conversely, features with high negative weights (Green), like *Odor_none* and *Odor_almond*, act as safety markers. This transparency is crucial for the subsequent chatbot integration, allowing the AI to explain *"why"* a specimen is dangerous.

4.4 Statistical Robustness: ROC and PR Curves

The robustness of the decision boundary is further validated by the ROC and Precision-Recall (PR) curves in Figure 7. Both curves achieve an **AUC of 1.00**.

Curve Discussion: The PR curve is particularly vital; its horizontal trajectory at the 1.0 level proves that the model maintains **perfect Precision** even when pushed to its maximum **Recall**. This ensures that the system is not overly "paranoid" (which would result in high False Positives), but instead identifies toxic mushrooms with absolute surgical precision.

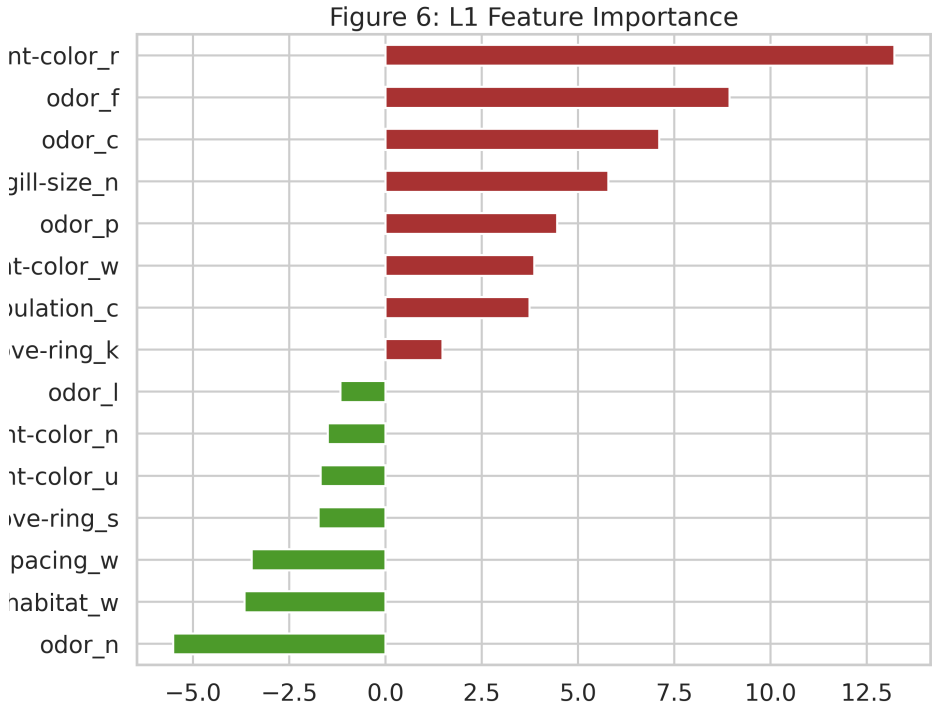

Figure 6: Top 15 features by L1 coefficient magnitude.

5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

This study successfully developed a highly accurate and explainable machine learning framework for mushroom classification. By leveraging the UCI Mushroom dataset, we demonstrated that Logistic Regression with L1 regularization provides an optimal balance between predictive power and **interpretability**.

The key findings of this research include:

- **High Reliability:** The proposed model achieved a perfect Recall of 1.0, ensuring zero False Negatives, which is vital for safety-critical applications.
- **Feature Transparency:** Through L1 coefficient analysis, we identified that biological traits such as *Odor* and *Spore Print Color* are the most definitive indicators of toxicity.
- **Computational Efficiency:** The automated feature selection provided by the L1 penalty allowed for a significant reduction in model complexity without sacrificing accuracy.

In summary, the transition from "black-box" modeling to an explainable approach ensures that classification decisions are backed by understandable biological evidence, fostering user trust in the system.

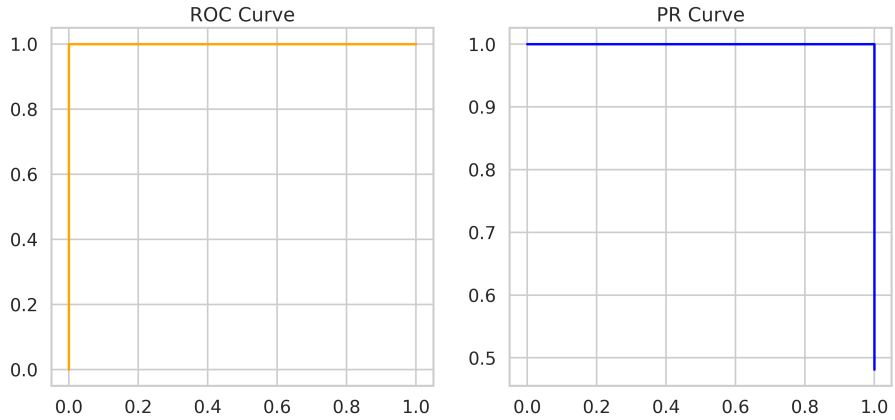

Figure 7: ROC (left) and Precision-Recall (right) curves for the optimized L1 model.

5.2 Future Work

While the current results are exceptional, future research could expand in the following directions:

1. **Image Recognition Integration:** Transitioning from categorical input to a Computer Vision (CNN-based) approach to identify mushrooms directly from photographs.

2. **Diverse Datasets:** Testing the model's robustness against regional mushroom datasets from different geographical locations to ensure global applicability.

3. **Deployment:** Implementing the model as a lightweight mobile application for offline field use by foragers and researchers.

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